Cross Spectra and Coherence

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Key Concepts

1) The cross spectrum describes the correlation between two stationary signals as a function of frequency.

    a) If the cross spectrum is large at a particular frequency, those components in the processes tend to be correlated. Positive values imply positive correlation while negative values imply negative correlation.

    b) If the cross spectrum is zero at a particular frequency, then components at that frequency in the processes are uncorrelated.

2) The magnitude-squared coherence is the magnitude squared of the cross spectrum normalized by the power spectra of each signal.

    a) The magnitude squared coherence between two processes is like the magnitude square correlation coefficient as a function of frequency.

    b) The magnitude squared coherence is always between zero and one.

3) The magnitude squared coherence is often used to test for the presence of a linear relationship between random signals.

4) The size of the magnitude squared coherence reflects the degree of relationship between the signals.

    a) If the magnitude squared coherence is near unity, then the signals at that frequency are related by a linear time-invariant system.

    b) If the magnitude squared coherence is small or zero, then the signals at that frequency are uncorrelated or unrelated.

Video Screenshots

The cross spectrum describes the correlation between two stationary signals in the frequency domain

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Cross correlation

$$r_{xy}[k] = E\{x[n]y^*[n-k]\} \quad \overset{DTFT}{\longleftrightarrow}$$

Cross Spectrum

$$S_{xy}(\omega) = \sum_{k=-\infty}^{\infty} r_{xy}[k] e^{-jk\omega}$$

Example: Fourier component

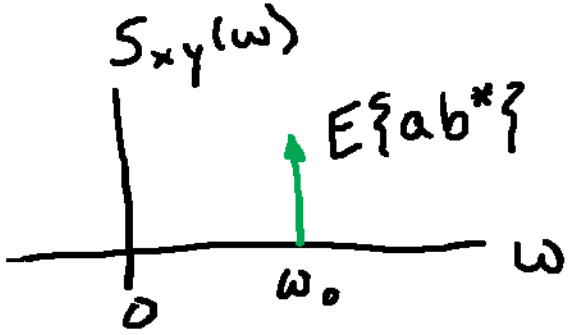
$x[n] = a e^{j\omega_0 n}$ $\quad y[n] = b e^{j\omega_0 n}$ $\quad$ a, b are random vars

$$r_{xy}[k] = E\{x[n]y^*[n-k]\} = E\{a e^{j\omega_0 n} b^* e^{-j\omega_0(n-k)}\}$$

$$= E\{ab^*\} e^{j\omega_0 k}$$

$$S_{xy}(\omega) = E\{ab^*\}\,\delta(\omega-\omega_0)$$

(0 for a, b independent)



Magnitude Squared Coherence (MSC)

- frequency domain analog of the correlation coefficient in statistics

$$C_{xy}(\omega) = \frac{|S_{xy}(\omega)|^2}{S_x(\omega)\, S_y(\omega)}$$

$$0 \le C_{xy}(\omega) \le 1$$

$(S_x(\omega), S_y(\omega) \neq 0)$

Fourier Component Example:

$x[n] = a e^{j\omega_0 n}$, $y[n] = b e^{j\omega_0 n}$ $\Longrightarrow$ $S_{xy}(\omega) = E\{ab^*\}\delta(\omega-\omega_0)$

$S_x(\omega) = E\{|a|^2\}\delta(\omega-\omega_0)$, $S_y(\omega) = E\{|b|^2\}\delta(\omega-\omega_0)$

$$C_{xy}(\omega) = \frac{|E\{ab^*\}|^2}{E\{|a|^2\}E\{|b|^2\}}, \quad \omega = \omega_0$$

(correlation coefficient)

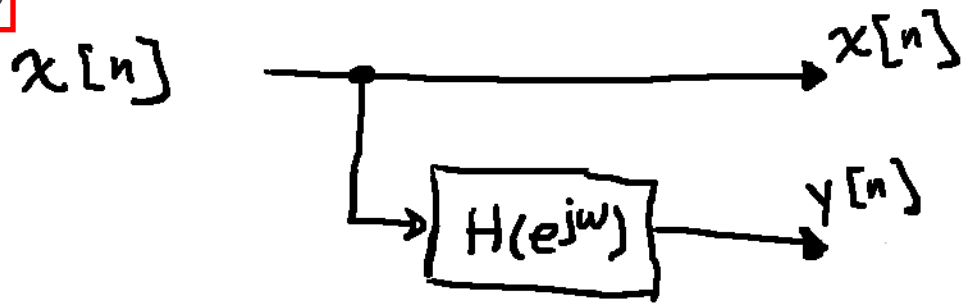
$C_{xy}(\omega_0) = 0 \Leftrightarrow$ a, b indep.

$C_{xy}(\omega_0) = 1 \Leftrightarrow$ a, b linearly dependent

e.g., $b = 2a$

MSC and LTI systems

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$$y[n] = \sum_{m=-\infty}^{\infty} h[m]\, x[n-m]$$

$$r_{xy}[k] = E\{x[n]\, y^*[n-k]\}$$

$$S_{xy}(\omega) = H^*(e^{j\omega})\, S_x(\omega)$$

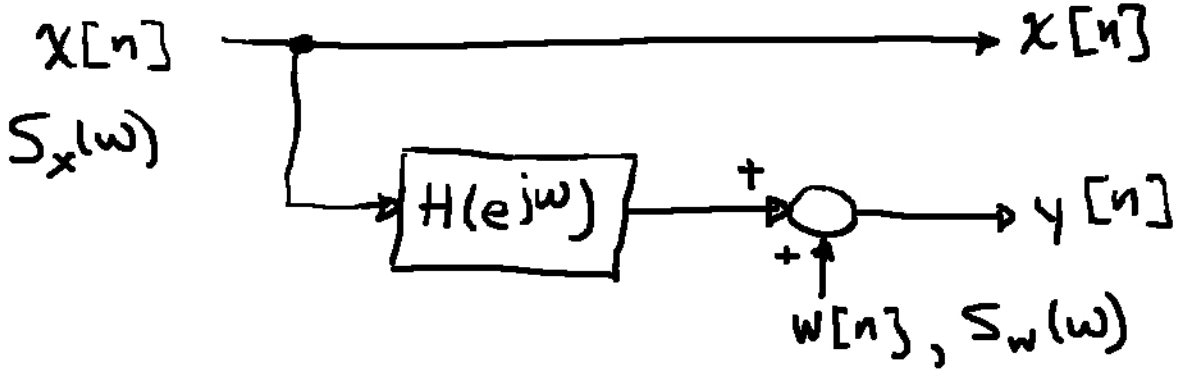
$$S_y(\omega) = |H(e^{j\omega})|^2 S_x(\omega)$$

$$C_{xy}(\omega) = \frac{|H^*(e^{j\omega})\, S_x(\omega)|^2}{S_x(\omega)\;\; |H(e^{j\omega})|^2 S_x(\omega)} = \frac{|H(e^{j\omega})|^2\; S_x^2(\omega)}{S_x^2(\omega)\;\; |H(e^{j\omega})|^2} = 1,$$

$$S_y(\omega) \neq 0$$

Random signals related through linear systems have unity magnitude squared coherence

Testing relationships between signals



$x[n]$, $w[n]$ independent

$$S_y(\omega) = |H(e^{j\omega})|^2 S_x(\omega) + S_w(\omega)$$

$$S_{xy}(\omega) = H^*(e^{j\omega}) S_x(\omega)$$

$$C_{xy}(\omega) = \frac{|H(e^{j\omega})|^2 S_x^2(\omega)}{S_x(\omega)\left[|H(e^{j\omega})|^2 S_x(\omega) + S_w(\omega)\right]} = \frac{|H(e^{j\omega})|^2 S_x(\omega)}{|H(e^{j\omega})|^2 S_x(\omega) + S_w(\omega)}$$

$$= \frac{\frac{|H(e^{j\omega})|^2 S_x(\omega)}{S_w(\omega)}}{1 + \frac{|H(e^{j\omega})|^2 S_x(\omega)}{S_w(\omega)}}$$

- $|H(e^{j\omega})|^2 S_x(\omega) \gg S_w(\omega) \Rightarrow C_{xy}(\omega) \approx 1$
- $|H(e^{j\omega})|^2 S_x(\omega) \ll S_w(\omega) \Rightarrow C_{xy}(\omega) \approx 0$
- $|H(e^{j\omega})|^2 S_x(\omega) \approx S_w(\omega) \Rightarrow C_{xy}(\omega) \approx 0.5$

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